Systems Biology Project 2 Report

Spatiotemporal Signal Propagation

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Task A1: Mutation Comparison

This section compares the Wild Type (WT) control with several specific mutations (`AKT1_E17K`, `PIK3CA_E545K`, `PIK3CA_H1047R`, `PTEN_del`) to evaluate their impact on spatiotemporal ERK signaling. The Relative Risk (RR) metric was used to quantify the likelihood of a cell exhibiting an ERK activity spike given that a neighboring cell recently spiked.

A Mann-Whitney U test with Bonferroni correction was performed to compare the relative risk of each mutant against the WT. The results are summarized in Table 1.

Mutation	Mean RR	Std Error	p-value vs WT	Significant
`AKT1_E17K`	1.581	0.072	3.46×10^{-3}	True
`PIK3CA_E545K`	1.707	0.032	0.184	False
`PIK3CA_H1047R`	3.200	0.131	1.20×10^{-5}	True
`PTEN_del`	1.564	0.082	3.46×10^{-3}	True

Table 1: Mutation comparison for spatiotemporal signal propagation relative to WT.

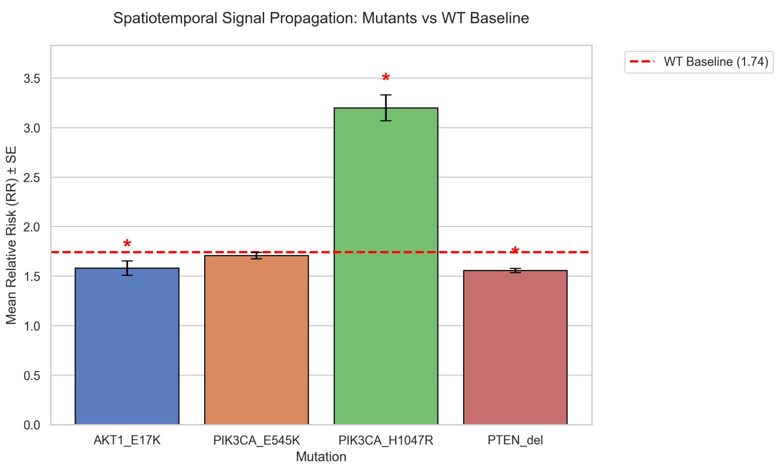

Figure 1: Mean Relative Risk (RR) $\pm$ SE across mutant cell lines compared to the WT baseline. Asterisks indicate statistical significance evaluated via Mann-Whitney U test with Bonferroni correction.

Biological Interpretation: The `PIK3CA_H1047R` mutation significantly increases the Relative Risk compared to WT. This mutation acts at the plasma membrane, potentially promoting the secretion of paracrine factors or increasing sensitivity to neighbor-derived signals. This "hypersensitivity" facilitates efficient signal spread. In contrast, downstream mutations like

`AKT1_E17K` and `PTEN_del` generate high constitutive baseline activity. This internal noise triggers negative feedback loops, desensitizing cells to external cues, which leads to asynchronous signaling and the observed drop in Relative Risk.

Task A2: Lagged Exposure

To understand the temporal dynamics of intercellular communication, we analyzed the Relative Risk across different time lags (τ) ranging from 0 to 30 minutes (0 to 6 frames) for the WT and two representative mutations: `AKT1_E17K` and `PTEN_del`. The objective was to determine if constitutive activation alters the optimal communication lag.

Lag τ (min)	**Lag (frames)**	**WT RR**	`AKT1_E17K` **RR**	`PTEN_del` **RR**
0	0	1.756	1.611	1.573
5	1	1.664	1.568	1.479
10	2	1.563	1.541	1.351
15	3	1.455	1.513	1.230
20	4	1.366	1.494	1.137
25	5	1.294	1.474	1.062
30	6	1.234	1.452	1.023

Table 2: Relative Risk ($RR^{(\tau)}$) across different exposure lags for WT, `AKT1_E17K`, and `PTEN_del`.

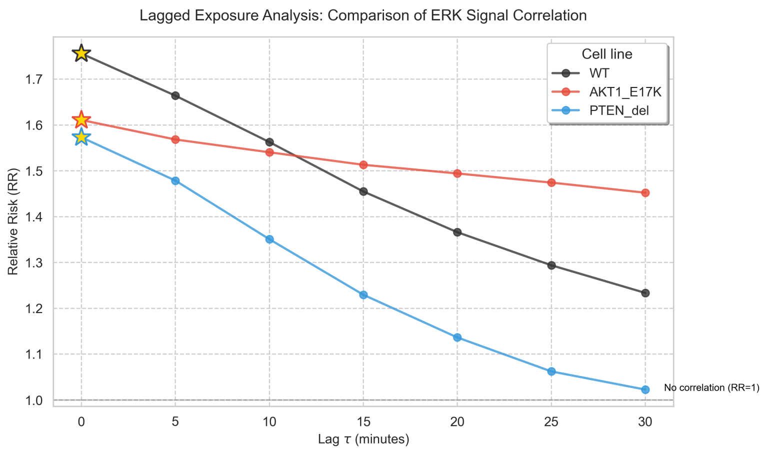

Figure 2: Temporal decay curves of lagged Relative Risk $RR^{(\tau)}$ over a 6-minute window for WT, `AKT1_E17K`, and `PTEN_del` cell lines.

Biological Interpretation: For all cell lines, the Relative Risk is highest at $\tau = 0$ and steadily decreases as the lag increases. The lack of a shifted peak indicates that signal transmission is extremely rapid, with cells responding almost immediately to a neighbor's signaling event. The convergence of RR toward 1 at longer lags confirms that the neighbor effect is transient. Furthermore, WT cells exhibit the strongest response ($RR \approx 1.76$). The reduced coupling strength in `AKT1_E17K` and `PTEN_del` reflects their constitutively elevated internal activity, reducing their sensitivity to external cues.

Task A3: Parameter Robustness

The robustness of the spatial radius parameter (r) was evaluated for the WT cell line. We tested radii of 30, 60, 90, and 150 μm to determine the optimal distance for capturing genuine local signaling events without introducing excessive background noise.

Radius (μm)	Exposed Nodes	P(exp)	P(unexp)	Relative Risk (RR)
30	43,171	0.180	0.087	2.068
60	146,620	0.131	0.075	1.756
90	239,638	0.112	0.070	1.605
150	332,701	0.100	0.075	1.328

Table 3: Impact of spatial radius r on Relative Risk in WT cells.

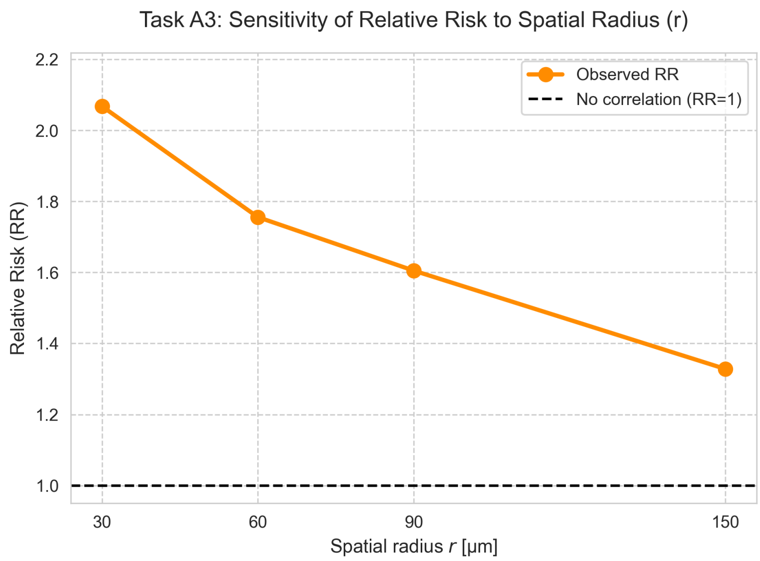

Figure 3: Sensitivity analysis showing the trade-off between Mean Relative Risk and average neighbor count across the spatial radius sweep.

Biological Interpretation: While a radius of 30 μm yields the highest Relative Risk ($RR \approx 2.07$), it captures very few neighbors per cell, leading to statistical volatility and the exclusion of genuine adjacent neighbors due to irregular cell shapes. Conversely, expanding the radius to 150 μm includes an enormous number of distant cells. This massive inclusion dilutes the measured effect as uncorrelated background noise overwhelms local signals (RR drops to 1.33). The chosen radius of 60 μm (spanning 1.5 to 2 epithelial cell diameters) is optimally aligned with the effective range of short-range paracrine diffusion, successfully capturing direct cell-to-cell communication.

Task B: Graph-based propagation decay analysis

0.1 Introduction

We investigated how ERK signaling propagation changes with increasing graph distance between cells.

Hypothesis: Propagation strength decreases with graph distance. The process differs between wild types and mutants.

Biological motivation: Mutations in the PI3K–AKT pathway may alter how signaling activity propagates through cellular communities. Studying how propagation strength changes may reveal whether mutant cell lines exhibit broader or more persistent spatial coordination compared to WT cells.

Research question: How does propagation strength decrease with graph distance? Does the decay differ in WT vs mutants?

0.2 Methods

For the analysis we selected one representative site for each mutation type:

```
MUTATIONS_SITES = {
    "WT": 1,
    "AKT1_E17K": 5,
    "PTEN_del": 17
}
```

These sites were used as representative examples of different signaling types observed across the dataset.

For each experiment-site block, we used the spatial neighbour graph generated in the original propagation pipeline.

To study how signaling propagation changes with distance, we analysed neighbours at increasing graph distances:

- distance 1 = direct neighbours,
- distance 2 = neighbours-of-neighbours,
- distance 3 = third-order neighbours.

For every cell, we checked whether at least one neighbour at distance d exhibited a jump event. We then measured the probability that the same cell performed a future jump afterwards. Propagation strength was quantified using relative risk (RR):

$$RR(d) = \frac{P(\text{future jump} \mid \text{exposed at distance } d)}{P(\text{future jump} \mid \text{not exposed at distance } d)}$$

0.3 Results

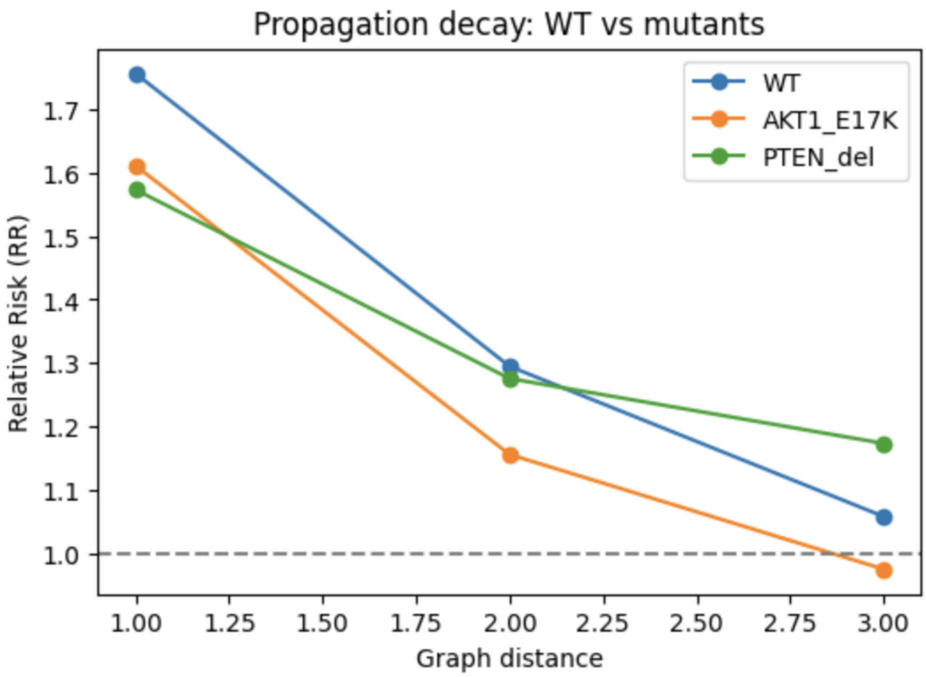

Figure 4: Propagation decay with graph distance in WT and mutant cell lines.

The analysis has shown, that relative risk of future jumps decreases with distance, indicating that signal propagation is strongest between nearby cells.

0.4 Discussion

Biological interpretation For first-degree neighbours, WT cells showed the strongest signal propagation. However, propagation decayed with distance, with RR close to baseline levels at graph distance 3.

AKT1_E17K cells showed weaker propagation overall. The decay with distance was the strongest, with RR falling below 1 at graph distance 3, suggesting loss of coordinated long-range propagation.

In case of PTEN_del cells, they had elevated RR values even at graph distance 3, showing more persistent spatial coordination.

These results suggest that oncogenic mutations affect not only the ERK activity, but also the range of signalling propagation.

Limitations: We were not able to analyze every site of every experiment due to computational complexity. Including all data and experiments could improve generalizability of the results.

Future directions Future work could extend the analysis to all experiments and compare multiple signaling pathways, such as ERK and FoxO simultaneously. It would also be interesting to investigate whether propagation patterns change under different environmental conditions.